Adhiyamaan College of Engineering (An Autonomous Institution), Hosur
Department of Computer Science and Engineering
An Academic Performance and Learning Management Web Application

ABSTRACT

This project presents an Academic Performance and Learning Management System (LMS) designed to provide a centralized digital platform for managing courses, assignments, assessments, attendance, and student analytics. Developed using a modern full-stack architecture with a secure backend and PostgreSQL database on Supabase, the system ensures reliable data management and role-based access control. The platform enables teachers to create structured assignments and assessments, while students interact through an intuitive dashboard. The system emphasizes performance monitoring through leaderboard-based rankings and visualization tools, improving efficiency, transparency, and academic decision-making.

OBJECTIVES

To develop a centralized Academic Performance and Learning Management System (LMS) that digitizes course management, assignments, assessments, attendance tracking, and performance analytics through a secure and scalable full-stack architecture.

METHODOLOGIES

Requirement Analysis & Problem Identification:
The development process began with a detailed analysis of existing academic management practices within educational institutions. Traditional systems were studied to identify limitations such as manual record maintenance, fragmented data storage, evaluation delays, and lack of performance visibility. Functional and non-functional requirements were defined to ensure the system addresses usability, security, scalability, and efficiency. This phase established the need for a centralized LMS capable of integrating assignments, assessments, attendance, and analytics.

System Design & Architecture Planning:
A modular full-stack system architecture was designed to separate the frontend, backend, and database layers. This approach ensured scalability, maintainability, and secure data flow. The frontend interface was structured to provide intuitive dashboards for Admin, Teachers, and Students. The backend was planned to handle authentication, business logic, validation, and API communication. A PostgreSQL relational database hosted on Supabase was selected for reliable data persistence and structured academic records management.

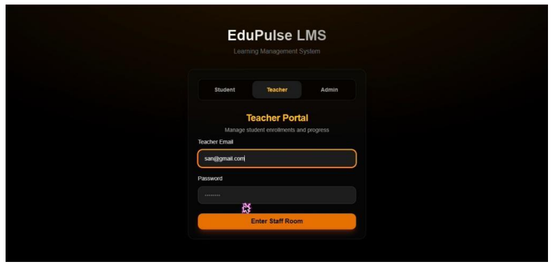

fig 1 : Profile page

Database Design & Data Modeling:
The database schema was designed using relational modeling principles to ensure consistency and integrity of academic data. Core entities such as Users, Courses, Assignments, Assessments, Attendance, Marks, Submissions, and Leaderboard metrics were defined. Relationships between these entities were established to support efficient querying and reporting. Constraints and validation rules were incorporated to maintain data accuracy, prevent redundancy, and ensure reliable performance tracking.

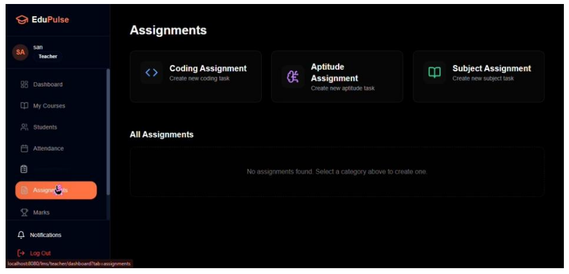

fig 2 : Assignment Module Interface

Visualization & Analytics Integration:
To enhance academic monitoring, leaderboard-based analytics and visualization mechanisms were implemented. Statistical computations and ranking algorithms were connected to visual components such as charts and summaries. These analytics enable teachers to interpret student performance, identify learning gaps, and monitor progress trends. Visualization tools transformed raw academic data into meaningful insights for decision-making.

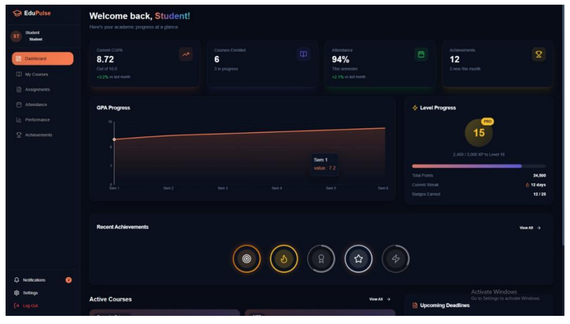

fig 3 : Visualization & Analytics- Student Profiler

Deployment & System Validation:
The final system configuration was validated through integration testing and real-scenario simulation. Backend services, database connectivity, and frontend dashboards were verified for consistency and correctness. The deployed system demonstrated stable performance, structured workflows, and accurate analytics generation.

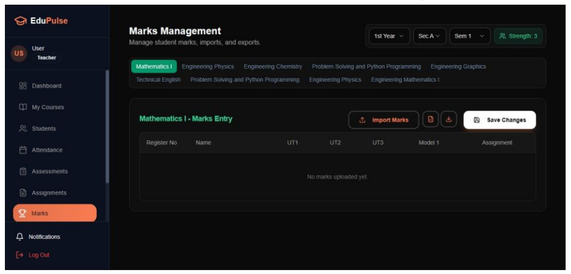

fig 4 : Marks Import -Teacher profile

ARCHITECTURE DIAGRAM

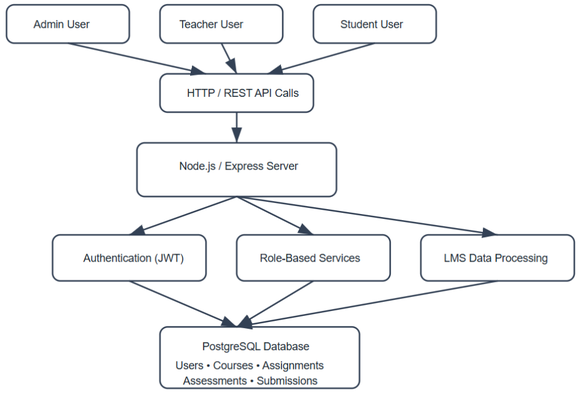

fig 5 : Module Architecture

RESULT

The Academic Performance and LMS platform delivered a structured and responsive digital environment with stable performance across modules. Role-based access, assignments, assessments, and leaderboard analytics functioned seamlessly, ensuring consistent interaction. Database optimization improved response times and maintained reliable academic records. Visualization and ranking mechanisms operated smoothly, enabling clear performance insights. System testing and validation confirmed stable operations, low latency, and accurate data processing, demonstrating the platform's reliability and efficiency.

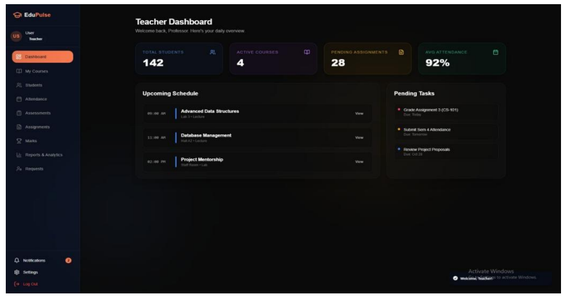

fig 6 : Teacher Dashboard

CONCLUSION

The Academic Performance and LMS platform demonstrated a reliable and scalable solution for digitizing academic workflows. The system ensured stable performance, efficient data management, and seamless interaction across assignments, assessments, attendance, and analytics modules. Gamified elements such as leaderboard rankings enhanced student engagement, motivation, and performance awareness. Visualization mechanisms provided meaningful academic insights, while database-driven operations maintained record accuracy and consistency. Overall, the platform proved effective in improving efficiency, transparency, usability, and engagement, making it suitable for modern educational environments and future enhancements.

REFERENCES

- Siemens, G. & Long, P., *"Penetrating the Fog: Analytics in Learning and Education"*, *EDUCAUSE Review*, 2011.
- Romero, C. & Ventura, S., *"Educational Data Mining: A Review of the State of the Art"*, *IEEE Transactions on Systems, Man, and Cybernetics C*, 2010.
- Baker, R. S. J. d., *"Data Mining for Education"*, *International Encyclopedia of Education*, 2010.
- Arnold, K. E. & Pistilli, M. D., *"Learning Analytics: Measuring and Understanding Student Learning in Higher Education"*, *EDUCAUSE Review*, 2012.

GUIDED BY

Mrs.C.Akshaya ME.,

TEAM MEMBERS

RASHAD AHMED.Z - 6176AC22UCS122
SAN.S - 6176AC22UCS125
SANJAY.M - 6176AC22UCS129